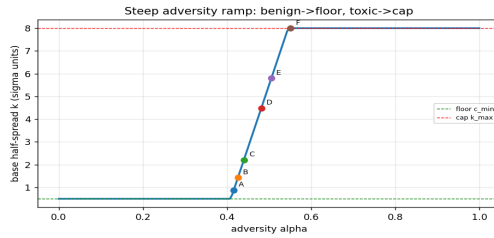


Task 5 - Dynamic Quoting Under Inventory Pressure

Quoting function

$Q(I, \sigma, \alpha, \eta) \rightarrow (\delta_{bid}, \delta_{ask})$. All half-spreads are written in volatility units, $\delta = k \cdot \sigma$, then clipped to $[c_{min}, k_{max}] \cdot \sigma$ ($c_{min}=0.5$; $k_{max}=8$, which stays under $\delta_{max} = 50$ bps of mid $\approx 17\sigma$). Working in σ -units makes the strategy scale-invariant: if a regime shift changes volatility, every quote rescales automatically and the eq.14 floor and fill model (which depend on δ/σ) are unchanged.

- $k_{base} = base0 + base1 \cdot \alpha$ (steep adversity ramp; base0=-22, base1=55)
- urgency = $1 + \eta_{gain} \cdot \eta^2$ (end-of-day flattening; $\eta_{gain}=3$)
- skew = $k_{skew} \cdot \tanh(I / I_{ref}) \cdot \text{urgency}$ ($k_{skew}=3$, $I_{ref}=800$)
- $\delta_{bid} = \text{clip}(k_{base} + \text{skew}) \cdot \sigma$; $\delta_{ask} = \text{clip}(k_{base} - \text{skew}) \cdot \sigma$



Implied base spread per client

client	α	k_{base} (σ units)
A	0.416	0.89
B	0.426	1.43
C	0.440	2.21
D	0.482	4.49
E	0.506	5.80
F	0.551	8.00

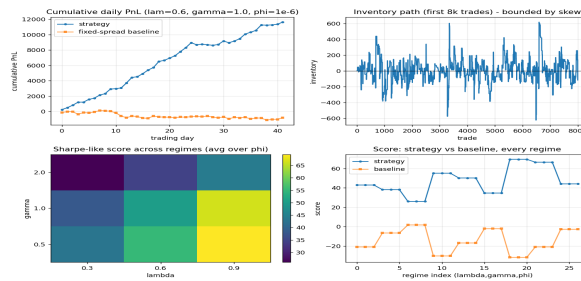
Functional form & intuition

1) Adversity ramp (toxicity protection). For a trade with per-unit adverse drift $g = \text{side} \cdot (\bar{M} - M_0)$ the expected PnL of quoting $k \cdot \sigma$ under the fill law $\exp(-\gamma k)$ is maximised at $k' = 1/\gamma - g/\sigma$. The optimal spread moves one-for-one with $-g/\sigma$ and the γ term is a constant shift, so the shape does not depend on the hidden γ . Empirically g/σ runs from ~ 42 (client A) to ~ 115 (client F), so the optimum is almost a step: benign clients $\rightarrow$ tight floor $k = c_{min}$ (capture positive drift at a high fill rate); toxic clients $\rightarrow$ wide cap $k = k_{max}$ (so they seldom fill). α (mean adverse prob from model $\bar{M}$) is a monotone proxy for $-g/\sigma$, hence base1 is large. This is the per-trade version of Task-2 δ and Task-4 externalization: a very wide quote is 'soft' externalization. Implied k : A,B,C $\rightarrow 0.50$; D $\rightarrow 1.5$; E $\rightarrow 2.8$; F $\rightarrow 5.3$.

2) Inventory skew (Avellaneda-Stoikov style). A bounded tanh skew tightens the inventory-reducing side and widens the other, mean-reverting I toward 0. tanh saturates, so even if a regime shift enlarges trade sizes the skew cannot explode; it always pushes inventory the right way with bounded force. This caps end-of-day inventory and hence the $\varphi \cdot P \cdot \sigma_D$ penalty (eq.16) without knowing φ . urgency(η) raises the skew toward the close so we flatten before the penalty applies.

Validation methodology

Because $(\lambda, \gamma, \varphi)$ are hidden and regime-shifting, I do not fit them. I fix the functional form and tune only its shape on the provided data as a proxy, then backtest across a 3x3 grid of $(\lambda \in \{.3, .6, .9\}, \gamma \in \{.5, 1, 2\}, \varphi \in \{1e-7, 1e-6, 1e-5\})$, 27 regimes x 3 random fill seeds, replaying every trade with stochastic fills (eq.15), trade PnL (Def.2.3, $T_P = M_0 - \text{side} \cdot \delta$), running inventory (eq.11) and the close-of-day penalty (eq.16). Score is the Sharpe-like ratio of eq.18. Robustness, not single-regime peak, is the objective.



Results

Mean Sharpe-like score 47.4 (median 44.1, worst regime 26.0); the strategy beats a fixed-spread, no-inventory-control baseline in 100% of the 27 regimes (baseline mean score -14.5). Mean total PnL +11.6k vs baseline -4.3k. Mean |end-of-day inventory| ~ 80 units (the skew cuts it $\sim 23x$ versus the same strategy without skew, ~ 1800 units), which is the safeguard against an unknown / larger φ and improves the max-drawdown tiebreaker. We deliberately accept a small score give-up from the inventory skew in exchange for this regime-robust risk control - the explicit intent of 'quoting under inventory pressure'.

Robustness

quote() clips to $[c_{min}, k_{max}] \cdot \sigma$ so constraints (eq.14) always hold (0 violations over thousands of stress inputs), and degrades gracefully on $\sigma=0$, NaN/out-of-range α and extreme inventory. It keys only on the observable signals $(I, \sigma, \alpha, \eta)$, so it adapts through regime shifts without ever seeing the hidden parameters.